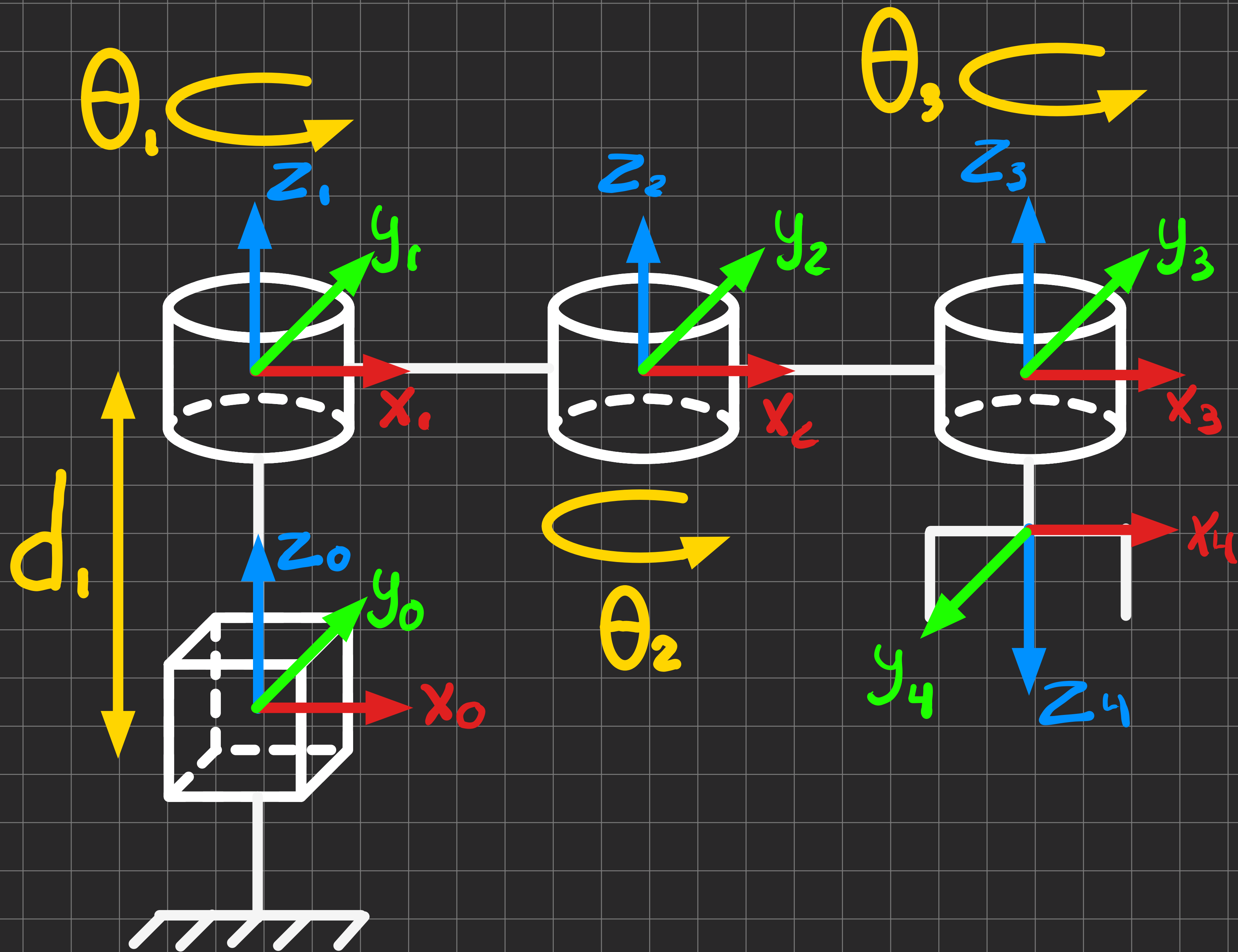


Cinematica Scara



Parametros DH

	a	α	θ	d	
l_1	0	0	0	q_1	$q_1 = d_1$
l_2	a_2	0	q_2	0	$q_2 = \theta_1$
l_3	a_3	0	q_3	0	$q_3 = \theta_2$
l_4	0	π	q_4	$-d_4$	$q_4 = \theta_3$

Matrices

$$A_1 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & q_1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$A_2 = \begin{bmatrix} \cos q_2 & -\sin q_2 & 0 & a_2 \cos q_2 \\ \sin q_2 & \cos q_2 & 0 & a_2 \sin q_2 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$A_3 = \begin{bmatrix} \cos q_3 & -\sin q_3 & 0 & a_3 \cos q_3 \\ \sin q_3 & \cos q_3 & 0 & a_3 \sin q_3 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$A_4 = \begin{bmatrix} \cos q_4 & \sin q_4 & 0 & 0 \\ \sin q_4 & -\cos q_4 & 0 & 0 \\ 0 & 0 & -1 & -d_4 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$A_{12} = \begin{bmatrix} \cos q_2 & -\sin q_2 & 0 & a_2 \cos q_2 \\ \sin q_2 & \cos q_2 & 0 & a_2 \sin q_2 \\ 0 & 0 & 1 & q_1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$A_{13} = \begin{bmatrix} \cos q_{23} & -\sin q_{23} & 0 & a_2 \cos q_2 + a_3 \cos q_{23} \\ \sin q_{23} & \cos q_{23} & 0 & a_2 \sin q_2 + a_3 \sin q_{23} \\ 0 & 0 & 1 & q_1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$A_{14} = \begin{bmatrix} \cos q_{24} & -\sin q_{24} & 0 & a_2 \cos q_2 + a_3 \cos q_{23} \\ \sin q_{24} & \cos q_{24} & 0 & a_2 \sin q_2 + a_3 \sin q_{23} \\ 0 & 0 & -1 & q_1 - d_4 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$q_{23} = q_1 + q_2$$

$$q_{24} = q_{23} + q_4$$